

Bethel Hall

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RESEARCH INTERESTS

Large Language Models, Pretraining & Post-Training, Formal Verification, Neurosymbolic AI, Trustworthy AI Systems

EDUCATION

Stevens Institute of Technology

Ph.D. in Computer Science; GPA: 3.92/4.00; Advisor: Prof. William Eiers

Expected May 2028

Hoboken, NJ

Addis Ababa University

BSc in Software Engineering

June 2022

RESEARCH EXPERIENCE

Graduate Research Assistant

Stevens Institute of Technology, Department of Computer Science

Sep 2022–Present

Hoboken, NJ

- **VeriFin.** Built an LLM verification system for numerical reasoning over SEC filings using XBRL grounding and Z3-based verification; resolved 98.8% of XBRL Filing cases without accepting an incorrect claim.
- **CloudFix.** Led development of an automated policy-repair system combining SMT fault localization, targeted LLM generation, and formal verification; outperformed an LLM-only baseline across all tested request sizes and released a 282-policy benchmark.
- **VeriMed.** Designed a neurosymbolic system that translates natural-language specifications into SMT constraints and uses solver-generated counterexamples for repair; achieved 98.5% verified accuracy.
- **PolicySummarizer.** Combined finite-state automata, LLM-based simplification, and model counting to generate verified policy characterizations; achieved 93% accuracy on the hardest evaluation task.

University of Michigan, Undergraduate Research Assistant

Developed safety-aware neural network training methods by incorporating formally specified constraints into the learning objective.

2021

SELECTED PUBLICATIONS

Hall et al. CloudFix: Automated Policy Repair for Cloud Access Control Policies Using Large Language Models. SANER, 2026.

Vatsa, Hall, and Eiers. Neurosymbolic Characterization for Reliable Access Control Policy Analysis. ISSRE, 2026.

Hall et al. VeriFin: A Neurosymbolic Framework for Verifying LLM-Generated Financial Claims. arXiv, 2026.

Hall and Eiers. Neurosymbolic Auditing of Natural-Language Software Requirements. arXiv, 2026.

ADDITIONAL EXPERIENCE

Teaching: Machine Learning, Java Programming

Reviewer: NeurIPS 2026, ICAIF 2026

TECHNICAL SKILLS

Languages: Python, Java, SQL, R | **ML & LLMs:** PyTorch, Hugging Face Transformers, vLLM, PEFT, TRL

LLM Training: Pretraining, Post-Training, Supervised Fine-Tuning

Verification & Reasoning: Z3, SMT/SMT-LIB, Model Counting

Systems & Data: AWS IAM, XBRL, SEC EDGAR